

Course Planner 2.0

From waiting days for an advisor to a valid course plan in minutes.

ThunderHacks 2026 · Main Track · Version 2.0

aryano96.github.io/LayerLabs

01 · Name the Version 1.0

Today: plan by hand, then email an advisor

Every Algoma student does this, every term.

Academic calendar
prerequisites

Degree requirements
a different chapter

Self-Service timetable
behind student login

→

The student cross-checks by hand
Which courses can I take, and will they fit?

→

Email an advisor
and wait 2–3 business days

02 · Show what is broken

Four specific problems

2–3
business days
The reply time on Algoma's own course-registration support page.

3
places to check
Prerequisites, degree rules and section times each live somewhere different.

243
combinations
Five courses with three sections each, checked for clashes by eye.

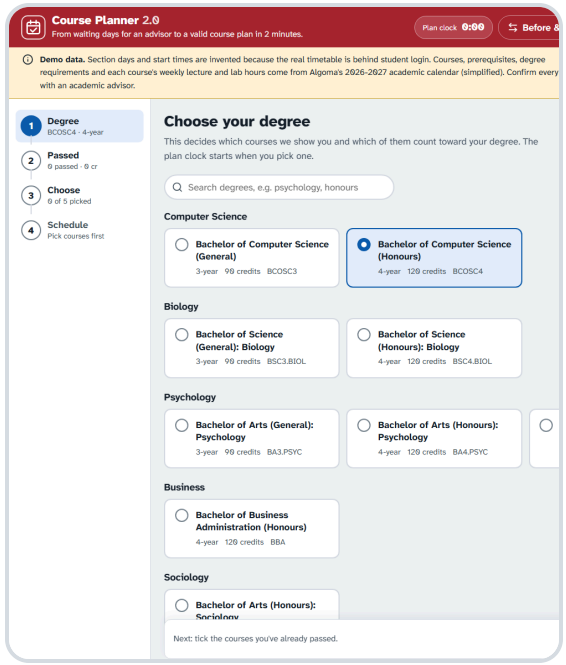
Rules written for people, not checked for them

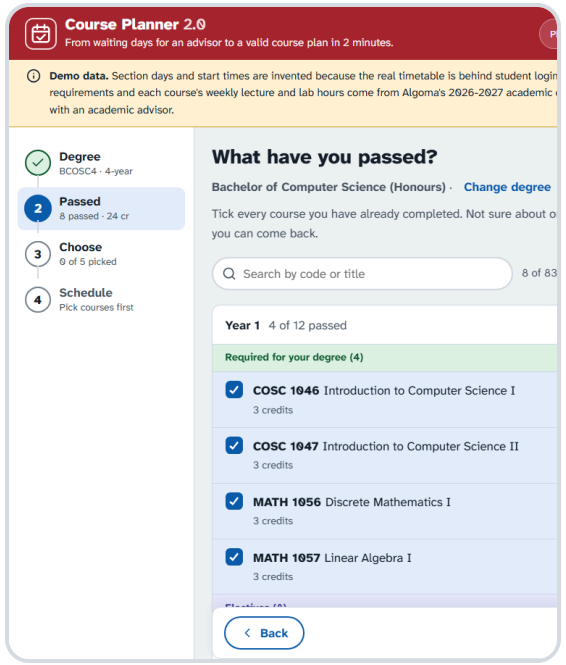
"Completion of second year core program in Computer Science or permission of the Department."

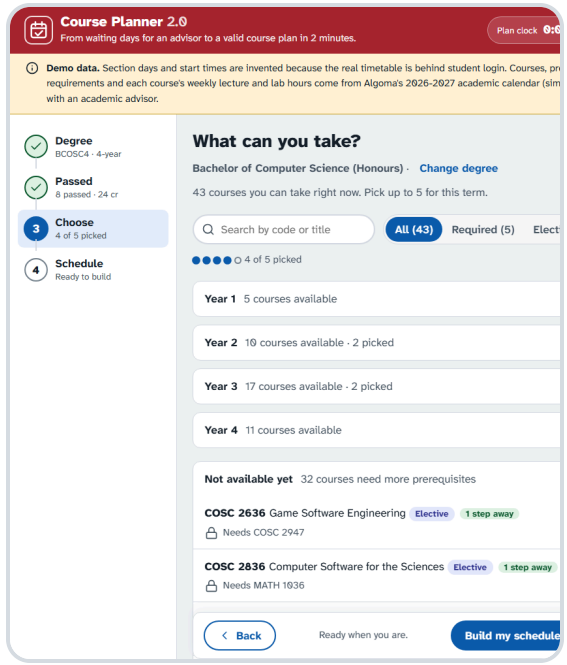
COSC 3506 Software Engineering, academic calendar

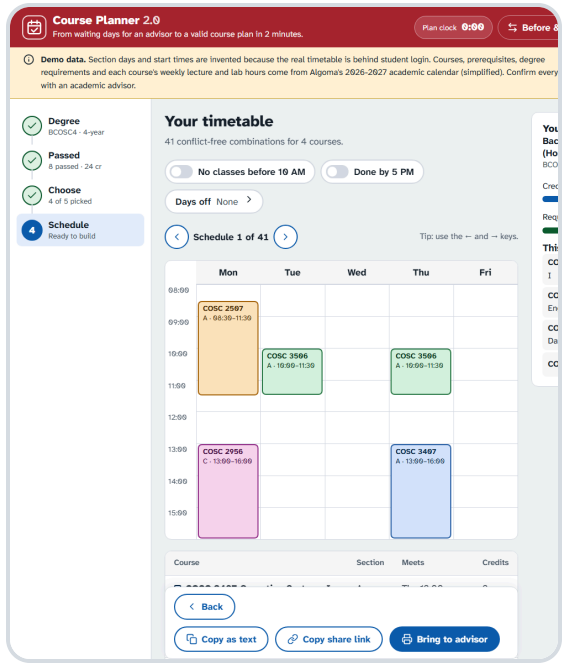
03 · Build the Version 2.0

Four steps from degree to timetable









1 Choose your degree

2 Tick what you have passed

3 See what you can take, and why not

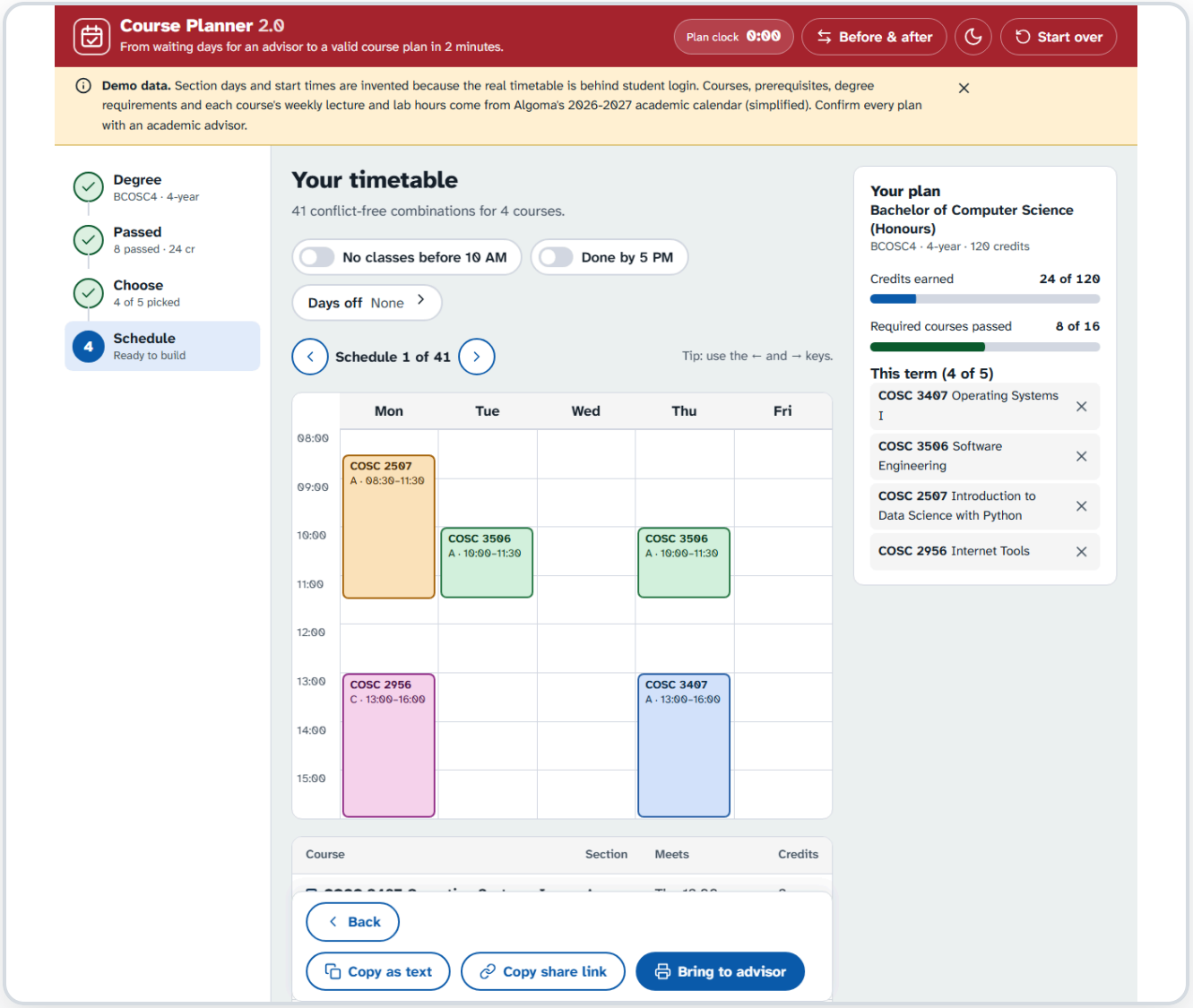
4 Get clash-free timetables

9 degrees · 307 courses · real prerequisites from the 2026–2027 academic calendar

Live demo

One student, one term

1. Pick **Computer Science (Honours)**
2. Tick the 8 courses already passed
3. See Required courses first
4. **COSC 3106** is “1 step away”: it needs MATH 2056
5. Pick 4 courses: **41** clash-free timetables
6. Filter early mornings, late days, Fridays



04 · Prove it is better

Before and after

	Today (Version 1.0)	Course Planner 2.0
Time	2–3 business days for a reply	**Minutes, measured by an on-screen plan clock**
Steps	8 manual steps	**4 steps**
Errors	Mistakes found by the advisor, days later	**No ineligible picks, no clashes: both covered by automated tests**
Access	Calendar PDFs and a login portal	**Any device, keyboard only, screen-reader table, light and dark**

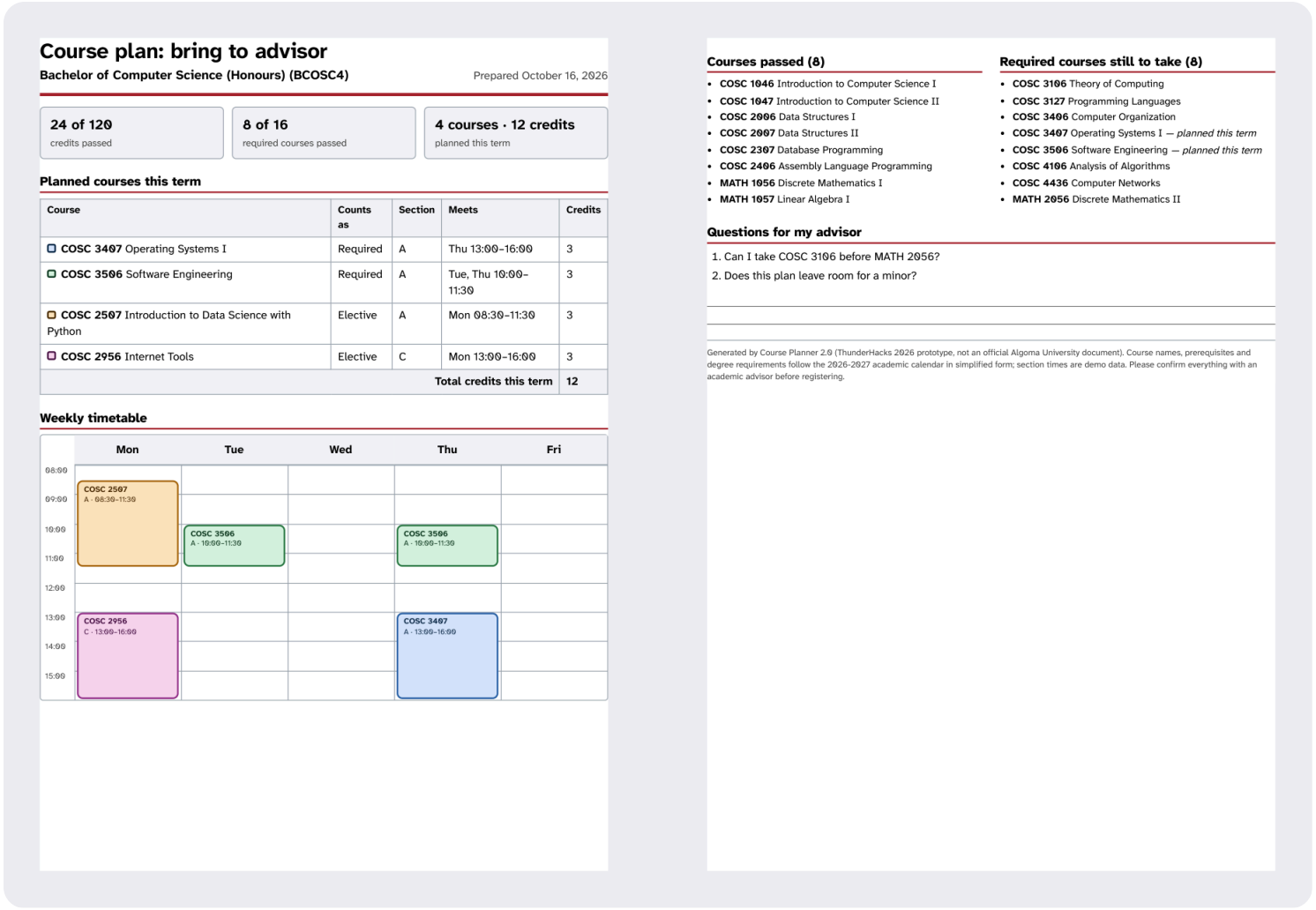
The 8 steps are the team’s walkthrough of today’s process.

The advisor's side

A confirmed plan, not a blank page

- ✓ Progress in credits and required courses
- ✓ The planned courses and weekly timetable
- ✓ What is still required
- ✓ The student's own questions

The advisor confirms a plan instead of building one.



Course plan: bring to advisor

Bachelor of Computer Science (Honours) (BCOSC4) Prepared October 16, 2026

24 of 120	8 of 16	4 courses · 12 credits
credits passed	required courses passed	planned this term

Planned courses this term

Course	Counts as	Section	Meets	Credits
COSC 3407 Operating Systems I	Required	A	Thu 13:00–16:00	3
COSC 3506 Software Engineering	Required	A	Tue, Thu 10:00–11:30	3
COSC 2507 Introduction to Data Science with Python	Elective	A	Mon 08:30–11:30	3
COSC 2956 Internet Tools	Elective	C	Mon 13:00–16:00	3
			Total credits this term	**12**

Weekly timetable

	Mon	Tue	Wed	Thu	Fri
08:00					
09:00	COSC 2507 A · 08:30–11:30				
10:00		COSC 3506 A · 10:00–11:30		COSC 3506 A · 10:00–11:30	
11:00					
12:00					
13:00	COSC 2956 C · 13:00–16:00			COSC 3407 A · 13:00–16:00	
14:00					
15:00					

Courses passed (8)

- **COSC 1046** Introduction to Computer Science I
- **COSC 1047** Introduction to Computer Science II
- **COSC 2006** Data Structures I
- **COSC 2007** Data Structures II
- **COSC 2307** Database Programming
- **COSC 2406** Assembly Language Programming
- **MATH 1056** Discrete Mathematics I
- **MATH 1057** Linear Algebra I

Required courses still to take (8)

- **COSC 3106** Theory of Computing
- **COSC 3127** Programming Languages
- **COSC 3406** Computer Organization
- **COSC 3407** Operating Systems I — *planned this term*
- **COSC 3506** Software Engineering — *planned this term*
- **COSC 4106** Analysis of Algorithms
- **COSC 4436** Computer Networks
- **MATH 2056** Discrete Mathematics II

Questions for my advisor

1. Can I take COSC 3106 before MATH 2056?
2. Does this plan leave room for a minor?

Generated by Course Planner 2.0 (ThunderHacks 2026 prototype, not an official Algoma University document). Course names, prerequisites and degree requirements follow the 2026-2027 academic calendar in simplified form; section times are demo data. Please confirm everything with an academic advisor before registering.

Real data · accessible · ready to switch

Built to be used after the weekend

Real data

Courses, prerequisites, credits and degree rules come from the 2026–2027 calendar. Only section days and start times are invented, and every screen says so.

Accessible

Keyboard only, visible focus, a high-legibility typeface, strong contrast, and a text table beside the timetable for screen readers.

Trustworthy

46 automated rule tests and over 100 browser checks. No server, no accounts, nothing leaves the browser.

Ready to switch

Students open a link. The registrar updates two data files each term. Another school swaps those files and it works unchanged.

A wait of days, reduced to minutes.

Try it, and read the code.

Live app **aryano96.github.io/LayerLabs**

Code **github.com/aryano96/LayerLabs**

Thank you · Questions welcome